

Summer Bootcamp – Session 1 Notes

Linear Algebra

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Purpose of the Session

This first three-hour session is a refresher on the linear algebra needed later in the bootcamp. The goal is not to cover linear algebra exhaustively, but to make students operational with the matrix language used in financial models, statistics, optimization, and numerical computation.

These notes are more detailed than the slides. They are organized by conceptual blocks, with definitions, propositions, and short proofs whenever a result is used later:

- algebraic operations;
- associated scalar and matrix operators;
- inversion and linear systems;
- injectivity, surjectivity, image, kernel, and rank;
- classical decompositions, including diagonalization and Cholesky.

1 Algebraic Operations

Definition 1.1 (Matrix and matrix-vector map). A matrix is a rectangular array of real numbers,

$$A = (a_{ij})_{1 \leq i \leq n, 1 \leq j \leq m} \in \mathcal{M}_{n,m}(\mathbb{R}).$$

The convention used here is that n is the number of rows and m is the number of columns. A matrix in $\mathcal{M}_{n,m}(\mathbb{R})$ maps vectors in $\mathbb{R}^m$ to vectors in $\mathbb{R}^n$.

Numerical example

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \in \mathcal{M}_{2,3}(\mathbb{R}), \quad x = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} \in \mathbb{R}^3.$$

The product Ax is defined because A has three columns and x has three entries. The result belongs to $\mathbb{R}^2$:

$$Ax = \begin{pmatrix} 1 \cdot 7 + 2 \cdot 8 + 3 \cdot 9 \\ 4 \cdot 7 + 5 \cdot 8 + 6 \cdot 9 \end{pmatrix} = \begin{pmatrix} 50 \\ 122 \end{pmatrix}.$$

Definition 1.2 (Addition and scalar multiplication). For two matrices of the same size,

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (\lambda A)_{ij} = \lambda a_{ij}.$$

Addition and scalar multiplication are entrywise operations.

Proposition 1.1 (Entrywise operations behave linearly). For matrices of the same size and scalars $\lambda, \mu \in \mathbb{R}$,

$$A + B = B + A, \quad (A + B) + C = A + (B + C),$$

$$\lambda(A + B) = \lambda A + \lambda B, \quad (\lambda + \mu)A = \lambda A + \mu A.$$

Proof. All identities are checked entry by entry. For instance,

$$(\lambda(A + B))_{ij} = \lambda(a_{ij} + b_{ij}) = \lambda a_{ij} + \lambda b_{ij} = (\lambda A + \lambda B)_{ij}.$$

The other identities follow from the same identities in $\mathbb{R}$. □

Example: addition and scalar multiplication

Let

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 & -1 \\ -3 & 1 & 5 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} 3 & -2 & 2 \\ -3 & 5 & 4 \end{pmatrix},$$

and

$$A - 2B = \begin{pmatrix} 1 - 4 & -2 - 0 & 3 - (-2) \\ 0 - (-6) & 4 - 2 & -1 - 10 \end{pmatrix} = \begin{pmatrix} -3 & -2 & 5 \\ 6 & 2 & -11 \end{pmatrix}.$$

Example: a linear combination of vectors

Let

$$u = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad v = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}.$$

Then

$$3u - 2v = 3 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} - 2 \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ -6 \end{pmatrix}.$$

This is the same vector-space operation used with the columns of a matrix.

Definition 1.3 (Matrix multiplication). If $A \in \mathcal{M}_{n,m}(\mathbb{R})$ and $B \in \mathcal{M}_{m,p}(\mathbb{R})$, then AB is defined and belongs to $\mathcal{M}_{n,p}(\mathbb{R})$. Its entries are

$$(AB)_{ij} = \sum_{k=1}^m a_{ik}b_{kj}.$$

The number of columns of A must equal the number of rows of B .

Dimension checks

Students should constantly track dimensions. Most beginner mistakes with matrix algebra are dimension mistakes:

$$(2 \times 3)(3 \times 1) \rightarrow 2 \times 1, \quad (4 \times 2)(2 \times 5) \rightarrow 4 \times 5,$$

whereas

$$(2 \times 3)(2 \times 5)$$

is not defined.

Proposition 1.2 (Matrix multiplication represents composition). Matrix multiplication represents composition of linear transformations. If B first maps a vector from $\mathbb{R}^p$ to $\mathbb{R}^m$, and A then maps from $\mathbb{R}^m$ to $\mathbb{R}^n$, the composed transformation is $AB : \mathbb{R}^p \rightarrow \mathbb{R}^n$.

Proof. For $x \in \mathbb{R}^p$, the i th coordinate of $(AB)x$ is

$$\sum_{j=1}^p (AB)_{ij}x_j = \sum_{j=1}^p \sum_{k=1}^m a_{ik}b_{kj}x_j = \sum_{k=1}^m a_{ik} \left(\sum_{j=1}^p b_{kj}x_j \right).$$

This is exactly the i th coordinate of $A(Bx)$. □

Worked example

Let

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 3 & 2 \end{pmatrix}.$$

Then

$$AB = \begin{pmatrix} 1 \cdot 2 + 2(-1) + (-1)3 & 1 \cdot 1 + 2 \cdot 0 + (-1)2 \\ 0 \cdot 2 + 3(-1) + 4 \cdot 3 & 0 \cdot 1 + 3 \cdot 0 + 4 \cdot 2 \end{pmatrix} = \begin{pmatrix} -3 & -1 \\ 9 & 8 \end{pmatrix}.$$

Matrix-vector product as a linear combination of columns

Let

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{pmatrix}, \quad x = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}.$$

Then

$$Ax = 1 \begin{pmatrix} 2 \\ 0 \end{pmatrix} - 2 \begin{pmatrix} -1 \\ 4 \end{pmatrix} + 0 \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ -8 \end{pmatrix}.$$

Thus Ax is a linear combination of the columns of A , with coefficients given by the entries of x .

Proposition 1.3 (Product rules). Matrix multiplication is associative and distributive whenever the dimensions are compatible. It is not commutative in general.

Proof. Associativity follows from the composition interpretation: $(AB)C$ and $A(BC)$ both send x to $A(B(Cx))$. Distributivity follows from linearity of matrix-vector maps, since both $A(B+C)$ and $AB+AC$ send x to $A(Bx+Cx)$. The example below proves that commutativity fails in general. $\square$

In general, $AB \neq BA$. For

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

we get

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}.$$

Thus this single example is enough to show that matrix multiplication is not commutative in general.

2 Associated Operators: Transpose, Trace, and Determinant

Definition 2.1 (Transpose). The transpose of $A = (a_{ij})$ is the matrix $A^\top$ satisfying

$$(A^\top)_{ij} = a_{ji}.$$

It turns rows into columns and columns into rows.

For example,

$$A = \begin{pmatrix} 1 & 3 & -2 \\ 0 & 4 & 5 \end{pmatrix} \implies A^\top = \begin{pmatrix} 1 & 0 \\ 3 & 4 \\ -2 & 5 \end{pmatrix}.$$

Proposition 2.1 (Transpose rules). Whenever the dimensions are compatible,

$$(A + B)^\top = A^\top + B^\top, \quad (\lambda A)^\top = \lambda A^\top, \quad (AB)^\top = B^\top A^\top.$$

For vectors x, y , the dot product can be written as $x^\top y$, and

$$(Ax)^\top y = x^\top A^\top y.$$

Proof. The first two identities are entrywise. For the product,

$$((AB)^\top)_{ij} = (AB)_{ji} = \sum_k a_{jk} b_{ki} = \sum_k (B^\top)_{ik} (A^\top)_{kj} = (B^\top A^\top)_{ij}.$$

The dot-product identity follows by applying this product rule to $(Ax)^\top y$. $\square$

Definition 2.2 (Symmetric and positive semidefinite matrices). A square matrix A is symmetric if $A^\top = A$. It is positive semidefinite if

$$x^\top A x \geq 0 \quad \text{for all } x \in \mathbb{R}^n.$$

It is positive definite if the inequality is strict for every nonzero x .

Proposition 2.2 ($X^\top X$ is positive semidefinite). For every real matrix X , the matrix $X^\top X$ is symmetric and positive semidefinite.

Proof. Symmetry follows from $(X^\top X)^\top = X^\top (X^\top)^\top = X^\top X$. For any vector z ,

$$z^\top X^\top X z = (Xz)^\top (Xz) = \|Xz\|^2 \geq 0.$$

Thus $X^\top X$ is positive semidefinite. □

For

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix},$$

we have

$$x^\top A x = 2x_1^2 + 2x_1x_2 + 2x_2^2 = (x_1 + x_2)^2 + x_1^2 + x_2^2.$$

This is strictly positive for $x \neq 0$, so A is positive definite.

Definition 2.3 (Trace). For a square matrix A , the trace is the sum of diagonal entries:

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}.$$

For example,

$$\text{tr} \begin{pmatrix} 3 & 1 & 0 \\ 2 & -4 & 5 \\ 1 & 1 & 7 \end{pmatrix} = 3 - 4 + 7 = 6.$$

Proposition 2.3 (Trace rules). Whenever the expressions are defined,

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B), \quad \text{tr}(\lambda A) = \lambda \text{tr}(A),$$

$$\text{tr}(AB) = \text{tr}(BA)$$

whenever both products make sense as square matrices.

Proof. Linearity is entrywise on the diagonal. If $A \in \mathcal{M}_{n,m}(\mathbb{R})$ and $B \in \mathcal{M}_{m,n}(\mathbb{R})$, then

$$\text{tr}(AB) = \sum_{i=1}^n \sum_{k=1}^m a_{ik} b_{ki} = \sum_{k=1}^m \sum_{i=1}^n b_{ki} a_{ik} = \text{tr}(BA).$$

□

Definition 2.4 (Determinant). The determinant is a scalar attached to a square matrix. It measures signed volume scaling and detects whether a square matrix loses dimension.

In $\mathbb{R}^2$, $|\det(A)|$ is the area scaling factor of the linear map $x \mapsto Ax$. In $\mathbb{R}^3$, $|\det(A)|$ is the volume scaling factor. If $\det(A) = 0$, some volume collapses to zero, so the transformation cannot be invertible.

Definition 2.5 (2×2 determinant). For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

Numerical examples

$$\det \begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix} = 2(-4) - 3(1) = -11.$$

The determinant is nonzero, so the matrix is invertible.

$$\det \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = 1 \cdot 4 - 2 \cdot 2 = 0.$$

The determinant is zero because the second row is twice the first. The rows do not span $\mathbb{R}^2$, and the matrix is not invertible.

Definition 2.6 (General determinant: the Leibniz formula). Let $A = (a_{ij}) \in \mathcal{M}_n(\mathbb{R})$ and let $\mathfrak{S}_n$ denote the set of all permutations of $\{1, \dots, n\}$. The determinant is

$$\det(A) = \sum_{\sigma \in \mathfrak{S}_n} \text{sgn}(\sigma) \prod_{i=1}^n a_{i, \sigma(i)}.$$

Here $\text{sgn}(\sigma) = 1$ for an even permutation and $\text{sgn}(\sigma) = -1$ for an odd permutation. Equivalently,

$$\text{sgn}(\sigma) = (-1)^{N(\sigma)}, \quad N(\sigma) = \#\{(i, j) : i < j, \sigma(i) > \sigma(j)\},$$

where $N(\sigma)$ is the number of inversions of σ .

Each term selects exactly one entry from every row and every column. Since $\mathfrak{S}_n$ contains $n!$ permutations, the formula has $n!$ signed terms. It is the clean general definition but a deliberately horrible computational method for large n ; elimination or a matrix factorization is far more efficient.

Definition 2.7 (Minors and cofactors). For $A \in \mathcal{M}_n(\mathbb{R})$, let A_{ij} be the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j . The corresponding minor and cofactor are

$$M_{ij} = \det(A_{ij}), \quad C_{ij} = (-1)^{i+j} M_{ij}.$$

The signs of the cofactors follow the checkerboard pattern

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

Proposition 2.4 (Laplace expansion along a row or a column). For any fixed row i ,

$$\det(A) = \sum_{j=1}^n a_{ij} C_{ij} = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}).$$

For any fixed column j ,

$$\det(A) = \sum_{i=1}^n a_{ij} C_{ij} = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}).$$

Thus one may expand along any row or any column. In practice, choose one that contains many zeros.

The 3×3 determinant

For

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix},$$

expansion along the first row gives

$$\det(A) = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}.$$

Equivalently,

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Numerical 3×3 example

Let

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \\ 2 & 0 & 4 \end{pmatrix}.$$

Expanding along the first row,

$$\det(A) = 1 \begin{vmatrix} 3 & 1 \\ 0 & 4 \end{vmatrix} - 2 \begin{vmatrix} -1 & 1 \\ 2 & 4 \end{vmatrix} + 0 \begin{vmatrix} -1 & 3 \\ 2 & 0 \end{vmatrix}.$$

Thus

$$\det(A) = 1(12) - 2((-1)4 - 1 \cdot 2) = 12 - 2(-6) = 24.$$

Since $\det(A) \neq 0$, this matrix is invertible.

Proposition 2.5 (Triangular determinant). If A is upper or lower triangular, then

$$\det(A) = \prod_{i=1}^n a_{ii}.$$

Proof. In the Leibniz formula for the determinant, every non-identity permutation chooses at least one entry below the diagonal for an upper triangular matrix (or above the diagonal for a lower triangular matrix). That entry is zero, so only the diagonal product remains. $\square$

For example,

$$\det \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{pmatrix} = 2 \cdot 3 \cdot (-2) = -12.$$

Proposition 2.6 (Determinant rules). For square matrices of the same size:

- $\det(I_n) = 1$.
- $\det(AB) = \det(A) \det(B)$.
- $\det(A^\top) = \det(A)$.
- If A is invertible, then $\det(A^{-1}) = 1/\det(A)$.
- Swapping two rows changes the sign of the determinant.
- Multiplying one row by λ multiplies the determinant by λ .
- Adding a multiple of one row to another row does not change the determinant.

Proof. The row-operation rules come from the determinant being multilinear in the rows and alternating: two equal rows force determinant zero, so swapping rows changes sign and adding a multiple of one row to another contributes only a determinant with two proportional rows. The identity and transpose rules follow from the Leibniz formula. The product rule $\det(AB) = \det(A) \det(B)$ follows because applying B and then A multiplies oriented volume first by $\det(B)$ and then by $\det(A)$. Once the product rule is known, $1 = \det(I_n) = \det(AA^{-1}) = \det(A) \det(A^{-1})$, which gives the inverse rule. $\square$

3 Inversion and Linear Systems

A linear system can be written as $Ax = b$. The matrix A contains coefficients, x contains unknowns, and b is the target vector.

Definition 3.1 (Identity and inverse). The identity matrix I_n satisfies $AI_n = A$ and $I_n A = A$ whenever dimensions are compatible. A square matrix A is invertible if there exists A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n.$$

Proposition 3.1 (2×2 invertibility and inverse formula). For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the matrix is invertible if and only if $\det(A) = ad - bc \neq 0$. In that case,

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Proof. If $ad - bc \neq 0$, direct multiplication of A by the displayed matrix gives I_2 on both sides. Conversely, if $ad - bc = 0$, the rows are linearly dependent, so a nonzero vector belongs to $\ker(A)$. Such a matrix cannot be invertible. $\square$

Proposition 3.2 (Invertible systems have one solution). If A is invertible, the system $Ax = b$ has the unique solution

$$x = A^{-1}b.$$

Proof. Multiplying $Ax = b$ by A^{-1} gives $x = A^{-1}b$, so any solution must be this vector. Conversely, $A(A^{-1}b) = b$, so this vector is indeed a solution. $\square$

For numerical work, however, solving $Ax = b$ directly is usually preferable to forming A^{-1} explicitly.

Example: solving a 2×2 system

Consider

$$\begin{cases} 2x_1 + 3x_2 = 5, \\ x_1 - 4x_2 = -2. \end{cases}$$

Then

$$A = \begin{pmatrix} 2 & 3 \\ 1 & -4 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ -2 \end{pmatrix}.$$

The determinant is

$$\det(A) = 2(-4) - 3(1) = -11 \neq 0,$$

so there is a unique solution. From the second equation, $x_1 = -2 + 4x_2$. Substituting into the first equation gives

$$2(-2 + 4x_2) + 3x_2 = 5 \quad \Rightarrow \quad 11x_2 = 9 \quad \Rightarrow \quad x_2 = \frac{9}{11}.$$

Then

$$x_1 = -2 + 4\frac{9}{11} = \frac{14}{11}.$$

Thus

$$x = \begin{pmatrix} 14/11 \\ 9/11 \end{pmatrix}.$$

Solving using the inverse

For a 2×2 matrix, the inverse formula gives

$$A^{-1} = \frac{1}{-11} \begin{pmatrix} -4 & -3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 4/11 & 3/11 \\ 1/11 & -2/11 \end{pmatrix}.$$

Then

$$x = A^{-1}b = \begin{pmatrix} 4/11 & 3/11 \\ 1/11 & -2/11 \end{pmatrix} \begin{pmatrix} 5 \\ -2 \end{pmatrix} = \begin{pmatrix} 14/11 \\ 9/11 \end{pmatrix}.$$

Proposition 3.3 (Forward substitution). If L is lower triangular with nonzero diagonal entries, the system $Lx = b$ can be solved uniquely from the first equation to the last. Similarly, an upper triangular system is solved by backward substitution.

Proof. The first equation contains only x_1 and has a unique solution because $\ell_{11} \neq 0$. Once $x_1, \dots, x_{i-1}$ are known, the i th equation has only one unknown left, namely x_i , and $\ell_{ii} \neq 0$ determines it uniquely. This induction solves all coordinates. $\square$

Triangular systems are solved sequentially. In fixed income, this is the linear algebra behind basic bootstrapping. Suppose

$$P_1 = 0.97, \quad P_2 = 0.92, \quad P_3 = 0.855.$$

Write

$$D(T_1) = y_1, \quad D(T_2) = y_1 + y_2, \quad D(T_3) = y_1 + y_2 + y_3.$$

Then

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0.97 \\ 0.92 \\ 0.855 \end{pmatrix}.$$

Forward substitution gives

$$y_1 = 0.97, \quad y_2 = 0.92 - 0.97 = -0.05, \quad y_3 = 0.855 - 0.92 = -0.065.$$

4 Injectivity, Surjectivity, Image, Kernel, and Rank

Definition 4.1 (Linear map associated with a matrix). A matrix $A \in \mathcal{M}_{m,n}(\mathbb{R})$ defines a linear map

$$T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad T_A(x) = Ax.$$

The notions of injectivity and surjectivity are properties of this map.

Definition 4.2 (Image and surjectivity). The image, or range, of A is

$$\text{Im}(A) = \{Ax : x \in \mathbb{R}^n\} \subset \mathbb{R}^m.$$

The map $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is surjective if every vector in $\mathbb{R}^m$ is reached:

$$\text{Im}(A) = \mathbb{R}^m.$$

Proposition 4.1 (Column span and surjectivity criterion). The image of A is the span of its columns. Therefore $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is surjective if and only if $\text{rank}(A) = m$.

Proof. Writing $A = [c_1 \ \cdots \ c_n]$ and $x = (x_1, \dots, x_n)^\top$ gives

$$Ax = x_1 c_1 + \cdots + x_n c_n.$$

Hence every vector in $\text{Im}(A)$ is in the span of the columns, and every linear combination of columns is equal to Ax for the vector of coefficients x . Surjectivity means this column span is all of $\mathbb{R}^m$, which is equivalent to rank m . $\square$

Definition 4.3 (Kernel and injectivity). The kernel, or null space, of A is

$$\ker(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

It is the set of directions collapsed by A .

The map $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is injective if different inputs give different outputs.

Proposition 4.2 (Kernel criterion for injectivity). For a linear map $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, injectivity is equivalent to

$$\ker(A) = \{0\}.$$

Equivalently, A has rank n .

Proof. If A is injective and $Ax = 0 = A0$, then $x = 0$, so $\ker(A) = \{0\}$. Conversely, if $\ker(A) = \{0\}$ and $Ax = Ay$, then $A(x - y) = 0$, hence $x - y = 0$ and $x = y$. By rank-nullity, $\ker(A) = \{0\}$ is equivalent to $\text{rank}(A) = n$. $\square$

Definition 4.4 (Rank). The rank of A is

$$\text{rank}(A) = \dim \text{Im}(A).$$

Proposition 4.3 (Rank-nullity theorem). The rank-nullity theorem says that for $A \in \mathcal{M}_{m,n}(\mathbb{R})$,

$$\text{rank}(A) + \dim \ker(A) = n,$$

where n is the number of columns of A .

Proof. Row-reduce A to echelon form. Row operations amount to multiplying on the left by an invertible matrix, so they preserve the kernel and the rank. If the echelon form has r pivot columns, then $r = \text{rank}(A)$ and the remaining $n - r$ variables are free. Therefore $\dim \ker(A) = n - r$, giving $\text{rank}(A) + \dim \ker(A) = r + (n - r) = n$. $\square$

Proposition 4.4 (Equivalent conditions for square matrices). For $A \in \mathcal{M}_n(\mathbb{R})$, injectivity, surjectivity, and invertibility are equivalent:

$$A \text{ injective} \iff A \text{ surjective} \iff A \text{ invertible}.$$

Equivalently:

- $\ker(A) = \{0\}$;
- $\text{Im}(A) = \mathbb{R}^n$;
- $\text{rank}(A) = n$;
- $\det(A) \neq 0$;
- for every $b \in \mathbb{R}^n$, $Ax = b$ has a unique solution.

Proof. For a square $n \times n$ matrix, rank-nullity shows that $\ker(A) = \{0\}$ is equivalent to $\text{rank}(A) = n$. Having rank n means the columns span $\mathbb{R}^n$, so $\text{Im}(A) = \mathbb{R}^n$. Thus injectivity, surjectivity, and full rank are equivalent. Full rank is equivalent to $\det(A) \neq 0$, hence to invertibility. If A is invertible, $Ax = b$ has the unique solution $A^{-1}b$; conversely, if every b has a unique solution, the columns form a basis and A is invertible. $\square$

Proposition 4.5 (Solution set of a linear system). For a general matrix $A \in \mathcal{M}_{m,n}(\mathbb{R})$:

- $Ax = b$ has at least one solution if and only if $b \in \text{Im}(A)$.
- If x_0 is one solution, then all solutions are

$$x = x_0 + z, \quad z \in \ker(A).$$

- If $\ker(A) = \{0\}$, then a solution, if it exists, is unique.

Proof. The first statement is exactly the definition of $\text{Im}(A)$. If x_0 solves $Ax = b$, then $A(x_0 + z) = b$ for every $z \in \ker(A)$. Conversely, if x is any solution, then $A(x - x_0) = 0$, so $x - x_0 \in \ker(A)$ and $x = x_0 + z$. If the kernel is $\{0\}$, this representation has no nonzero direction to add, so the solution is unique. $\square$

Definition 4.5 (Orthogonal vectors and orthogonal complement). Two vectors $x, y \in \mathbb{R}^m$ are orthogonal, written $x \perp y$, if

$$x^\top y = 0.$$

If $V \subset \mathbb{R}^m$ is a subspace, its orthogonal complement is

$$V^\perp = \{y \in \mathbb{R}^m : y^\top v = 0 \text{ for every } v \in V\}.$$

Thus $V^\perp$ contains exactly the directions perpendicular to the whole subspace V .

Proposition 4.6 (Fundamental orthogonality relation). For $A \in \mathcal{M}_{m,n}(\mathbb{R})$,

$$\text{Im}(A)^\perp = \ker(A^\top), \quad \text{Im}(A) = (\ker A^\top)^\perp.$$

Proof. For $y \in \mathbb{R}^m$, the condition $y \in \text{Im}(A)^\perp$ means $y^\top Ax = 0$ for every $x \in \mathbb{R}^n$. This is equivalent to $(A^\top y)^\top x = 0$ for every x , hence to $A^\top y = 0$. This proves $\text{Im}(A)^\perp = \ker(A^\top)$. The second identity follows by taking orthogonal complements in finite dimension. $\square$

Worked example: kernel and image

Let

$$A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

To find the kernel, solve $Ax = 0$:

$$\begin{cases} x_1 - x_2 = 0, \\ -x_1 + 2x_2 - x_3 = 0, \\ -x_2 + x_3 = 0. \end{cases}$$

The first and third equations give $x_1 = x_2$ and $x_3 = x_2$. Hence

$$x = t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \ker(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

Since the kernel is one-dimensional and A has three columns,

$$\text{rank}(A) = 3 - 1 = 2.$$

The image is a two-dimensional plane in $\mathbb{R}^3$.

Because A is symmetric, $A^\top = A$. Using the fundamental relation

$$\text{Im}(A) = (\ker A^\top)^\perp,$$

we get

$$\text{Im}(A) = (\ker A)^\perp = \{y \in \mathbb{R}^3 : y_1 + y_2 + y_3 = 0\}.$$

Numerical example: a rectangular matrix

Let

$$B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}.$$

The second row is twice the first, so the two rows are dependent. The columns are

$$c_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad c_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2c_1, \quad c_3 = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = 3c_1.$$

Thus

$$\text{Im}(B) = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}, \quad \text{rank}(B) = 1.$$

By rank-nullity,

$$\dim \ker(B) = 3 - \text{rank}(B) = 2.$$

Indeed, $Bx = 0$ means

$$x_1 + 2x_2 + 3x_3 = 0.$$

Taking $x_2 = s$ and $x_3 = t$ gives $x_1 = -2s - 3t$, so

$$\ker(B) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Here $B : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is neither injective nor surjective: it has a nontrivial kernel, and its image is only a line in $\mathbb{R}^2$.

5 Classical Decompositions

Why decompose matrices?

Matrix decompositions rewrite a matrix in a form that makes its action easier to understand or compute. A decomposition often answers one of the following questions:

- Can we solve a system faster?
- Can we understand the main directions of a transformation?
- Can we separate scaling, rotation, and correlation effects?
- Can we simulate or transform correlated random variables?

5.1 Eigenvalue Diagonalization

5.1.1 Definition and intuition

The simplest possible matrix is a diagonal matrix:

$$D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Its action is component-by-component:

$$Dx = \begin{pmatrix} \lambda_1 x_1 \\ \vdots \\ \lambda_n x_n \end{pmatrix}.$$

Diagonalization asks whether a general matrix can become diagonal after changing coordinates.

Definition 5.1 (Diagonalizable matrix). A matrix $A \in \mathcal{M}_n(\mathbb{R})$ is diagonalizable over $\mathbb{R}$ if there exist an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}.$$

Equivalently, in the coordinate system defined by P , the matrix A acts like the diagonal matrix D .

Definition 5.2 (Eigenvalue, eigenvector, eigenspace). An eigenvector of $A \in \mathcal{M}_n(\mathbb{R})$ is a nonzero vector v such that

$$Av = \lambda v.$$

The scalar λ is the associated eigenvalue. The direction of v is preserved by A ; only its magnitude and possibly its orientation change. The eigenspace associated with λ is

$$E_\lambda = \ker(A - \lambda I).$$

Proposition 5.1 (Eigenbasis criterion). If there is a basis of eigenvectors $(v_1, \dots, v_n)$, then A is diagonalizable. Put these eigenvectors in the columns of P :

$$P = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_n \\ | & & | \end{pmatrix}, \quad D = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Then

$$AP = PD, \quad \text{hence} \quad A = PDP^{-1}.$$

Conversely, if $A = PDP^{-1}$ with D diagonal, the columns of P form a basis of eigenvectors of A .

Proof. If $Av_i = \lambda_i v_i$ for each basis vector, then the i th column of AP is Av_i and the i th column of PD is $\lambda_i v_i$. Hence $AP = PD$, and since P is invertible, $A = PDP^{-1}$. Conversely, if $A = PDP^{-1}$, then $AP = PD$. The i th column gives $Ap_i = \lambda_i p_i$, where p_i is the i th column of P . Since P is invertible, these columns form a basis. $\square$

Definition 5.3 (Characteristic polynomial). Eigenvalues are found by solving

$$\det(A - \lambda I) = 0.$$

The polynomial

$$p_A(\lambda) = \det(A - \lambda I)$$

is the characteristic polynomial.

Proposition 5.2 (Eigenvalues are roots of the characteristic polynomial). A real number λ is an eigenvalue of A if and only if $p_A(\lambda) = 0$.

Proof. The equation $Av = \lambda v$ with $v \neq 0$ is equivalent to $(A - \lambda I)v = 0$ having a nonzero solution. This happens if and only if $A - \lambda I$ is not injective, equivalently not invertible. For a square matrix, non-invertibility is equivalent to determinant zero. $\square$

5.1.2 Computing eigenvalues and eigenvectors

Let

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

Then

$$A - \lambda I = \begin{pmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{pmatrix}$$

and

$$\det(A - \lambda I) = (4 - \lambda)(3 - \lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 5)(\lambda - 2).$$

Thus $\lambda_1 = 5$ and $\lambda_2 = 2$.

For $\lambda_1 = 5$,

$$A - 5I = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix},$$

so $-v_1 + v_2 = 0$. We may choose

$$u_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For $\lambda_2 = 2$,

$$A - 2I = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix},$$

so $2v_1 + v_2 = 0$. We may choose

$$u_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

5.1.3 Computational purpose

Diagonalization turns matrix operations into scalar operations. If $A = PDP^{-1}$, then

$$A^k = PD^kP^{-1}, \quad D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k).$$

This is useful for powers of matrices, linear recurrences, Markov chains, discrete-time dynamical systems, covariance analysis, and PCA.

5.1.4 Algorithm

1. Compute $p_A(\lambda) = \det(A - \lambda I)$.
2. Find the eigenvalues.
3. For each eigenvalue λ , solve $(A - \lambda I)v = 0$.
4. Check whether the eigenvectors found form a basis of $\mathbb{R}^n$.
5. If yes, build P from eigenvectors and D from eigenvalues.

Proposition 5.3 (Distinct real eigenvalues imply diagonalizability). If $A \in \mathcal{M}_n(\mathbb{R})$ has n distinct real eigenvalues, then A is diagonalizable over $\mathbb{R}$.

Proof. Choose one eigenvector v_i for each distinct eigenvalue λ_i . Suppose $\sum_i \alpha_i v_i = 0$. For a fixed j , apply the polynomial $q_j(A) = \prod_{i \neq j} (A - \lambda_i I)$ to this relation. Every term vanishes except the j th one, giving

$$\alpha_j \prod_{i \neq j} (\lambda_j - \lambda_i) v_j = 0.$$

The product is nonzero and $v_j \neq 0$, so $\alpha_j = 0$. Thus the v_i are linearly independent. There are n of them, so they form an eigenbasis. $\square$

Definition 5.4 (Algebraic and geometric multiplicities). The algebraic multiplicity of an eigenvalue is its multiplicity as a root of the characteristic polynomial. The geometric multiplicity is the dimension of the eigenspace

$$E_\lambda = \ker(A - \lambda I).$$

Proposition 5.4 (Multiplicity criterion). Always,

$$1 \leq \dim E_\lambda \leq \text{algebraic multiplicity of } \lambda.$$

The matrix is diagonalizable if and only if the sum of the dimensions of all eigenspaces is n . Equivalently, when the characteristic polynomial splits over $\mathbb{R}$, A is diagonalizable if and only if for every eigenvalue the geometric multiplicity equals the algebraic multiplicity.

Proof. The eigenspace is nonzero by definition of eigenvalue, so its dimension is at least one. If $g = \dim E_\lambda$, choose a basis whose first g vectors span E_λ . In this basis, A has a block form with λI_g in the upper-left block, so the characteristic polynomial contains the factor $(\lambda - t)^g$ up to sign. Hence g is at most the algebraic multiplicity. Eigenspaces associated with distinct eigenvalues are linearly independent, as proved above. Therefore A is diagonalizable exactly when the eigenspaces together provide n independent eigenvectors, i.e. when their dimensions sum to n . $\square$

Proposition 5.5 (Immediate sufficient cases). Every diagonal matrix is diagonalizable. Every triangular matrix with distinct diagonal entries is diagonalizable.

Proof. For a diagonal matrix, the canonical basis vectors are eigenvectors. For a triangular matrix, the eigenvalues are the diagonal entries because $A - \lambda I$ is triangular and its determinant is the product of diagonal entries. If these diagonal entries are distinct, the previous proposition applies. $\square$

5.1.5 Worked example: a general two-by-two matrix

Use the previous matrix:

$$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}.$$

We found eigenvalues $\lambda_1 = 5$, $\lambda_2 = 2$, with eigenvectors

$$u_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

Set

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -2 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 2 \end{pmatrix}.$$

Since

$$\det(P) = 1(-2) - 1(1) = -3 \neq 0,$$

P is invertible. Hence

$$A = PDP^{-1}.$$

Here

$$P^{-1} = \frac{1}{-3} \begin{pmatrix} -2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{pmatrix}.$$

This gives an efficient formula for powers:

$$A^k = P \begin{pmatrix} 5^k & 0 \\ 0 & 2^k \end{pmatrix} P^{-1}.$$

5.1.6 Worked example: a triangular matrix

Let

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}.$$

Since A is triangular, its eigenvalues are the diagonal entries:

$$\lambda_1 = 3, \quad \lambda_2 = 2.$$

For $\lambda = 3$,

$$A - 3I = \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

so $v_2 = 0$ and we may choose

$$u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

For $\lambda = 2$,

$$A - 2I = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

so $v_1 + v_2 = 0$ and we may choose

$$u_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Thus

$$P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}, \quad A = PDP^{-1}.$$

5.1.7 Worked example: a covariance-style matrix

Let

$$C = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}.$$

The characteristic polynomial is

$$\det(C - \lambda I) = (1 - \lambda)^2 - 0.8^2.$$

Thus

$$1 - \lambda = \pm 0.8,$$

so

$$\lambda_1 = 1.8, \quad \lambda_2 = 0.2.$$

For $\lambda_1 = 1.8$, an eigenvector is $(1, 1)^\top$. For $\lambda_2 = 0.2$, an eigenvector is $(1, -1)^\top$. Normalize them:

$$q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Set

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 1.8 & 0 \\ 0 & 0.2 \end{pmatrix}.$$

Since Q is orthogonal, $Q^{-1} = Q^\top$. Therefore

$$C = QDQ^\top.$$

Interpretation: most variance lies in the common direction $(1, 1)$, while much less variance lies in the spread direction $(1, -1)$.

5.1.8 Counterexample: a non-diagonalizable matrix

Let

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The only eigenvalue is $\lambda = 1$, with algebraic multiplicity 2. But

$$J - I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

so

$$(J - I)v = 0 \implies v_2 = 0.$$

The eigenspace is

$$E_1 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\},$$

which has dimension 1, not 2. Therefore J is not diagonalizable.

5.1.9 Diagonalization over the reals versus the complex numbers

The rotation matrix

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

has characteristic polynomial

$$\det(R - \lambda I) = \lambda^2 + 1.$$

It has no real eigenvalues, so it is not diagonalizable over $\mathbb{R}$. It is diagonalizable over $\mathbb{C}$, with eigenvalues i and $-i$. This distinction matters whenever we specify the field over which diagonalization is being performed.

5.1.10 Application: a discrete-time dynamical system

Consider the recursion

$$x_{k+1} = Ax_k, \quad A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}, \quad x_0 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The triangular matrix used in the slides was diagonalized as

$$A = PDP^{-1}, \quad P = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}.$$

Since

$$P^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

we obtain, for every integer $k \geq 0$,

$$A^k = PD^kP^{-1} = \begin{pmatrix} 3^k & 3^k - 2^k \\ 0 & 2^k \end{pmatrix}.$$

Therefore the complete trajectory is

$$x_k = A^k x_0 = \begin{pmatrix} 3^k - 2^k \\ 2^k \end{pmatrix}.$$

Instead of iterating a coupled two-dimensional recursion, diagonalization separates the system into the two scalar growth modes 3^k and 2^k . The mode associated with the largest eigenvalue eventually dominates.

5.2 Spectral Decomposition

Proposition 5.6 (Spectral theorem). If A is real and symmetric, then it can be diagonalized using an orthogonal matrix:

$$A = QDQ^\top.$$

The eigenvalues are real, and eigenvectors associated with distinct eigenvalues are orthogonal.

Proof. The Rayleigh quotient $x^\top Ax$ on the unit sphere attains a maximum because the unit sphere is compact. At a maximizer q_1 , the Lagrange multiplier condition gives $Aq_1 = \lambda_1 q_1$, so q_1 is an eigenvector. If $y \perp q_1$, then $q_1^\top Ay = (Aq_1)^\top y = \lambda_1 q_1^\top y = 0$, so the orthogonal complement of q_1 is invariant under A . Applying the same argument by induction gives an orthonormal eigenbasis. Placing these vectors in the columns of Q gives $Q^\top Q = I$ and $A = QDQ^\top$. $\square$

This theorem is central for covariance matrices, PCA, and risk-factor decompositions.

5.2.1 Application: common and spread risk factors

Return to the covariance matrix

$$C = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}.$$

Its orthonormal eigenvectors and eigenvalues are

$$q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \lambda_1 = 1.8, \quad q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \lambda_2 = 0.2.$$

For any unit portfolio direction q , its return variance is $q^\top Cq$. Consequently,

$$q_1^\top Cq_1 = 1.8, \quad q_2^\top Cq_2 = 0.2.$$

The first portfolio moves both assets in the same direction and represents a common factor. The second is a long-short spread. The spectral decomposition therefore produces orthogonal portfolios and measures exactly how much risk is carried by each one.

5.3 Cholesky Decomposition

Proposition 5.7 (Cholesky decomposition). If A is symmetric positive definite, then there exists a unique lower triangular matrix L with positive diagonal entries such that

$$A = LL^\top.$$

This is the Cholesky decomposition.

Proof. Proceed by induction on the dimension. Write

$$A = \begin{pmatrix} a & r^\top \\ r & B \end{pmatrix}, \quad a > 0.$$

Set $\ell_{11} = \sqrt{a}$ and $\ell = r/\sqrt{a}$. The Schur complement $S = B - \ell\ell^\top = B - rr^\top/a$ is positive definite: for $y \neq 0$, choose $\alpha = -r^\top y/a$ and use positive definiteness of $(\alpha, y)^\top A(\alpha, y)$. By induction, $S = MM^\top$ with M lower triangular and positive diagonal. Then

$$L = \begin{pmatrix} \sqrt{a} & 0 \\ \ell & M \end{pmatrix}$$

satisfies $A = LL^\top$. Uniqueness follows by matching entries from the first column onward; the positive diagonal fixes the sign at each step. $\square$

5.3.1 Worked Cholesky factorization

Let

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}.$$

Seek

$$L = \begin{pmatrix} \ell_{11} & 0 \\ \ell_{21} & \ell_{22} \end{pmatrix}$$

such that $A = LL^\top$. Then

$$LL^\top = \begin{pmatrix} \ell_{11}^2 & \ell_{11}\ell_{21} \\ \ell_{11}\ell_{21} & \ell_{21}^2 + \ell_{22}^2 \end{pmatrix}.$$

Matching entries gives

$$\ell_{11} = 2, \quad 2\ell_{21} = 2 \Rightarrow \ell_{21} = 1,$$

and

$$\ell_{21}^2 + \ell_{22}^2 = 3 \Rightarrow 1 + \ell_{22}^2 = 3 \Rightarrow \ell_{22} = \sqrt{2}.$$

Therefore

$$A = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & \sqrt{2} \end{pmatrix}.$$

5.3.2 Why Cholesky matters

Cholesky is widely used to:

- solve positive definite linear systems efficiently;
- simulate correlated Gaussian vectors;
- work with covariance matrices;
- implement numerical optimization algorithms.

If $Z \sim N(0, I)$ and $\Sigma = LL^\top$, then

$$X = LZ$$

has covariance matrix Σ .

5.3.3 Application: simulation of correlated returns

Let Z_1, Z_2 be independent standard Gaussian variables and take the factor computed above,

$$L = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix}.$$

Define

$$X = L \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix} = \begin{pmatrix} 2Z_1 \\ Z_1 + \sqrt{2}Z_2 \end{pmatrix}.$$

Since $\text{Cov}(Z) = I_2$,

$$\text{Cov}(X) = L \text{Cov}(Z) L^\top = LL^\top = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}.$$

Thus X_1 and X_2 have variances 4 and 3, covariance 2, and correlation

$$\rho_{12} = \frac{2}{\sqrt{4}\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

This gives a practical simulation recipe: generate independent standard Gaussian shocks and multiply them by L to impose the desired covariance structure.

6 Statistics and Finance Applications

Definition 6.1 (Empirical covariance matrix). If $X \in \mathbb{R}^{n \times d}$ is a centered data matrix, the empirical covariance matrix is

$$\hat{\Sigma} = \frac{1}{n-1} X^\top X.$$

Proposition 6.1 (Covariance matrices are positive semidefinite). The matrix $\hat{\Sigma}$ is symmetric and positive semidefinite.

Proof. Symmetry follows from $(X^\top X)^\top = X^\top X$. For every $z \in \mathbb{R}^d$,

$$z^\top \hat{\Sigma} z = \frac{1}{n-1} z^\top X^\top X z = \frac{1}{n-1} \|Xz\|^2 \geq 0.$$

□

Numerical covariance example

Let

$$X = \begin{pmatrix} 0.01 & -0.03 \\ 0.02 & 0.01 \\ -0.01 & 0 \end{pmatrix}.$$

Then

$$X^\top X = \begin{pmatrix} 0.0006 & -0.0001 \\ -0.0001 & 0.0010 \end{pmatrix}.$$

Definition 6.2 (Portfolio return). For asset returns r with covariance matrix Σ and portfolio weights w , the portfolio return is

$$R_p = w^\top r,$$

where the weights w_i represent the portfolio allocation.

Proposition 6.2 (Portfolio variance formula). The variance of the portfolio return is

$$\text{Var}(R_p) = w^\top \Sigma w.$$

Proof. Let $\mu = \mathbb{E}[r]$ and $\Sigma = \mathbb{E}[(r - \mu)(r - \mu)^\top]$. Then

$$\text{Var}(w^\top r) = \mathbb{E}[(w^\top (r - \mu))^2] = w^\top \mathbb{E}[(r - \mu)(r - \mu)^\top] w = w^\top \Sigma w.$$

□

For

$$\Sigma = \begin{pmatrix} 0.04 & 0.018 \\ 0.018 & 0.09 \end{pmatrix}, \quad w = \begin{pmatrix} 0.6 \\ 0.4 \end{pmatrix},$$

we compute

$$w^\top \Sigma w = 0.6^2(0.04) + 2(0.6)(0.4)(0.018) + 0.4^2(0.09) = 0.03744.$$

This quadratic form decomposes risk into individual variance terms and a covariance term.

Proposition 6.3 (PCA as covariance diagonalization). For centered data, PCA diagonalizes the covariance matrix

$$\hat{\Sigma} = \frac{1}{n-1} X^\top X.$$

The principal directions are eigenvectors of $\hat{\Sigma}$, and the corresponding eigenvalues are variances along those directions.

Proof. By the spectral theorem, $\hat{\Sigma} = QDQ^\top$ with Q orthogonal and D diagonal. If q_i is a column of Q , then the variance of the projected data Xq_i is

$$\frac{1}{n-1} \|Xq_i\|^2 = q_i^\top \hat{\Sigma} q_i = q_i^\top QDQ^\top q_i = D_{ii}.$$

Thus eigenvectors give the orthogonal directions of variation and eigenvalues give the amount of variance in those directions. $\square$

Let

$$X = \begin{pmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then

$$\hat{\Sigma} = \frac{1}{2} X^\top X = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

The eigenvalues are 2 and 0, with eigenvectors

$$q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

All the variance lies along the line $x_1 = x_2$.

Projection onto the first principal component

The first principal direction is $q_1 = (1, 1)^\top / \sqrt{2}$. The projected coordinates are

$$Xq_1 = \begin{pmatrix} -1 & -1 \\ 0 & 0 \\ 1 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sqrt{2} \\ 0 \\ \sqrt{2} \end{pmatrix}.$$

This keeps the direction where the data vary. Projection onto q_2 would give zero for every observation, because there is no variation in the orthogonal direction.

7 Markov Chains and Transition Matrices

7.1 From conditional probabilities to a matrix

Let $(X_n)_{n \geq 0}$ take values in a finite state space $\{1, \dots, m\}$. It is a Markov chain if the conditional distribution of the next state depends on the past only through the current state:

$$\mathbb{P}(X_{n+1} = i \mid X_n = j, X_{n-1}, \dots, X_0) = \mathbb{P}(X_{n+1} = i \mid X_n = j).$$

Definition 7.1 (Transition matrix). Using the column-vector convention of these notes, define

$$P_{ij} = \mathbb{P}(X_{n+1} = i \mid X_n = j).$$

Thus column j contains the probabilities of the next state given that the current state is j . Every entry is nonnegative and every column sums to one:

$$P_{ij} \geq 0, \quad \sum_{i=1}^m P_{ij} = 1.$$

Such a matrix is called column-stochastic.

Some books instead use row probability vectors and row-stochastic matrices. Both conventions are equivalent after transposition; the important point is to state the convention and use it consistently.

7.2 Distribution dynamics and powers of the matrix

Let

$$\mu_n = \begin{pmatrix} \mathbb{P}(X_n = 1) \\ \vdots \\ \mathbb{P}(X_n = m) \end{pmatrix}.$$

By the law of total probability,

$$(\mu_{n+1})_i = \sum_{j=1}^m \mathbb{P}(X_{n+1} = i \mid X_n = j) \mathbb{P}(X_n = j) = \sum_{j=1}^m P_{ij} (\mu_n)_j.$$

Therefore

$$\boxed{\mu_{n+1} = P\mu_n}, \quad \boxed{\mu_n = P^n \mu_0}.$$

More generally, the entry $(P^n)_{ij}$ is the probability of being in state i after n steps when the chain starts from state j .

This is precisely why matrix powers matter in probability: multiplying transition matrices adds time steps.

7.3 A two-state credit example

Consider two credit states, Good (G) and Bad (B). From G, the rating stays Good with probability 0.9 and deteriorates with probability 0.1. From B, it recovers with probability 0.4 and stays Bad with probability 0.6. With columns indexed by the current state,

$$P = \begin{pmatrix} 0.9 & 0.4 \\ 0.1 & 0.6 \end{pmatrix}.$$

If the obligor starts in state G, then $\mu_0 = (1, 0)^\top$ and

$$\mu_1 = P\mu_0 = \begin{pmatrix} 0.9 \\ 0.1 \end{pmatrix}, \quad \mu_2 = P^2\mu_0 = \begin{pmatrix} 0.85 \\ 0.15 \end{pmatrix}.$$

Thus the two-year probability of being in the Bad state is 0.15.

7.4 Stationary distributions and diagonalization

Definition 7.2 (Stationary distribution). A probability vector π is stationary if

$$P\pi = \pi, \quad \mathbf{1}^\top \pi = 1, \quad \pi_i \geq 0.$$

It is therefore an eigenvector of P associated with eigenvalue 1, normalized so that its entries sum to one.

For the credit matrix,

$$P\pi = \pi \iff 0.1\pi_G = 0.4\pi_B, \quad \pi_G + \pi_B = 1,$$

so

$$\pi = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix}.$$

The eigenvalues of P are 1 and 0.5. Since the matrix is diagonalizable, its powers separate the persistent stationary mode from the transient mode. In fact, when $\mu_0 = (1, 0)^\top$,

$$\mu_n = \begin{pmatrix} 0.8 \\ 0.2 \end{pmatrix} + 0.2(0.5)^n \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Because $(0.5)^n \rightarrow 0$, the distribution converges to π . This illustrates a central link between linear algebra and probability:

eigenvalue 1 determines the long-run distribution, while eigenvalues of modulus less than one determine the speed at which transient effects disappear.